Komal Adhikari

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SUMMARY

Computer Science student at Texas State University passionate about artificial intelligence, trustworthy machine learning, and software engineering. Experienced in developing full-stack applications and independently exploring AI concepts through personal projects. I enjoy connecting theory with practical implementation and am seeking research opportunities where I can contribute, learn from experienced researchers, and continue building reliable and impactful AI systems.

EDUCATION

Texas State University, San Marcos TX **GPA- 3.83/4.0** **Exp. Graduation: May 2028**

Bachelor of Science in Computer Science, with a Minor in Mathematics.

Relevant Coursework: Data Structures & Algorithms, Data Visualization, Object-Oriented Programming, Calculus(1&2), Software Engineering

PROJECTS AND RESEARCH

Undergraduate Researcher | Physics-Informed AI for Aviation Visibility **June 2026- Present**

- Investigating physics-informed computer vision and backward reasoning for interpretable airport visibility estimation under foggy conditions.
- Applying weakest-precondition reasoning inspired by DeepInfer to derive physically meaningful preconditions for explaining model predictions.
- Evaluating the approach using the Foggy AirSim City Images (FACI) synthetic dataset to compare prediction performance and interpretability against a standard vision baseline.

Cloud Garage | Microservices Bottleneck Detection Platform *Angular, TypeScript, NgRx, Spring Boot, REST APIs, Git, Jira*

- Contributed to a team-built diagnostic platform that identifies performance bottlenecks across microservices and presents supporting evidence to help developers investigate system issues.
- Implemented NgRx state management for project workflows, including actions, reducers, effects, feature state, and unit tests for loading, success, failure, and reset states.
- Contributed to authentication and API gateway security design using Auth0, JWT validation, and Spring Security within a microservices architecture.
- Collaborated through Agile sprints, Jira, Git branching, pull requests, code reviews, and integration with independently developed frontend and backend services.

DAYate | Full-Stack Date Planning Platform *React, Node.js, Express, MongoDB, JWT*

- Designed and developed a full-stack application that helps users organize and share date plans with loved ones.
- Applied modular software architecture principles (SRP, OCP) to create scalable and maintainable backend services.

TECHNICAL SKILLS AND UNIQUE INTERESTS

- **Programming Languages:** Python, C++, C , JavaScript, HTML, CSS, Assembly, Java
- **Frameworks & Libraries:** React.js, Node.js, Express.js, Scikit-Learn, NumPy, Pandas,Angular, NgRx, SpringBoot
- **Developer Tools & Technologies:** Git, GitHub, VS Code, VisualVM, Figma, Canva, Jira
- **Technical Exposure:** Experience working with structured systems, debugging, and learning complex tools through programming projects

EXPERIENCE

Resident Assistant (Front Desk) **Aug 2025- Present**

- Managed front desk operations, ensuring accurate record-keeping and secure handling of resident information

Volunteer – TXST All Engineering and Technology Job and Internship Fair **Feb 2026**

- Engaged with industry professionals and gained exposure to engineering and technology career opportunities.

Newsletter Writer & Editor | Chautari Pathways – Substack **Aug 2026 – Present**

- Write and publish newsletter content highlighting education, mentorship, student experiences, and community issues relevant to Chautari's mission.

LEADERSHIP AND INVOLVEMENT

Chautari Pathways | Social Media & Outreach **Aug 2026 – Present**

- Create digital content and outreach campaigns for a free college mentorship program serving students in Nepal.
- Promote college-access resources and mentorship opportunities for Grade 12 and gap-year students.

Austin Women in Technology (AWT), Member **Jan 2025- Present**

- Participated in networking events, technical talks, and mentorship opportunities.

Nepali Students Association (NSA), Social Media Manager **Dec 2024 – Oct 2025**

- Managed and scaled social media presence by designing and executing content strategies, increasing community engagement and strengthening organizational visibility through targeted campaigns.